

JENNIFER ZAMUDIO, Ph.D. Candidate

University of Michigan, Ann Arbor — Industrial & Operations Engineering
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Research Interests: **Human Factors Engineering in Healthcare, Healthcare Systems Design, Patient-Clinician Interaction, Measurement Science, Cognitive Ergonomics, Implementation Science**

EDUCATION

University of Michigan, Ann Arbor

- Ph.D. in Industrial & Operations Engineering (Human Factors Engineering)

Aug. 2023 – Present (Expected Defense in Spring 2027)

Dissertation: “Understanding, assessing and measuring patient-provider connection”

Advisor: Dr. Xi Jessie Yang (Committee members: Dr. Sanjay Saint, Dr. Amy Cohn, Dr. Amelia Warden)

California State University, Long Beach

- B.A. in Psychology

Aug. 2015 – May 2020

Honors: Cum Laude, Ronald E. McNair Scholar

RESEARCH & PROFESSIONAL EXPERIENCE

University of Michigan, Ann Arbor

- Graduate Research Assistant, Department of Industrial & Operations Engineering

Aug. 2023 – Present

Advisor: Dr. X. Jessie Yang, Interaction & Collaboration Research Lab

- Lead patient-oriented human factors research in partnership with Michigan Medicine and the VA Ann Arbor Healthcare System (AHRQ-supported), examining how clinical system design shapes patient-clinician connection and well-being.
- Developing and validating a multidimensional patient-clinician connection scale using a rigorous mixed-methods approach: qualitative thematic analysis (NVivo), cognitive interviewing (think-aloud protocols), expert-patient alignment studies, and large-scale psychometric testing.
- Conducting construct validation using confirmatory factor analysis (CFA), differential item functioning (DIF), fit statistics, and convergence with established relational and communication measures.
- Translating patient-identified dimensions of connection into co-designed clinical guidelines with physicians to: (1) strengthen relational dimensions of care, (2) disrupt antecedents of disconnection, and (3) operationalize system-level redesign principles for implementation within specialty care settings.
- Conducted a national mixed-methods survey of internal medicine physicians examining how human factors engineering can improve clinician well-being and patient-facing time, identifying structural barriers that constrain meaningful patient interaction.

- Research Program Coordinator (Rackham Aspiring Fellows), Rackham Graduate School

May 2024 – Present

Supervisor: Richard Nunn, Partnerships for Access, Community, and Excellence (PACE)

- Co-lead year-long faculty-student mentorship initiative supporting MSI scholars in research career development, facilitating workshops on research design, translational impact, and graduate preparation.
- Redesigning research undergraduate program structure, curriculum, and engagement model to improve trainee support and mentorship infrastructure.

Cedars-Sinai Medical Center, Los Angeles, CA

- Clinical Research Associate, Department of Surgery – Human Factors / Systems Engineering

May 2021 – Jul. 2023

Advisor: Dr. Tara N. Cohen,

- Conducted observational research (160+ hours) in robotic-assisted surgery across three academic medical centers to evaluate workflow disruptions, team dynamics, and patient safety outcomes.

- Applied HFACS-Healthcare and flow disruption taxonomies to classify adverse events and system vulnerabilities using adverse event report data.
- Analyzed intraoperative workload across surgical team roles and examined contributing factors to intraoperative mortality through structured chart review.
- Designed and implemented a gamified simulation-based training intervention (“Robotic Olympics”) across Cedars-Sinai and MUSC, adapting the program to site-specific organizational constraints (implementation science approach).
- Conducted physician-centered usability studies for medical device optimization to improve safety and reduce cognitive workload.

California State University, Long Beach

- Research Intern in McNair Scholars Summer Research Program, Department of Psychology

Jun. 2019 – Aug. 2019

Advisor: Dr. Gabriella Hancock, Stress & Technology Applied Research Lab

- Examined heart rate variability as a physiological marker of cognitive workload, contributing to human performance assessment methods.

- Research Assistant, Department of Psychology

Jun. 2018 – Jun. 2019

Advisor: Dr. Arturo Zavala, Addictions Lab

- Investigated behavioral and neurobiological mechanisms of substance use using experimental animal models.

- Research Lab Coordinator, Department of Psychology

Jun. 2017 – July 2018

Advisor: Dr. Sherry A. Span, Span Lab

- Examined distress tolerance as a moderator between ADHD symptoms and alcohol misuse using quantitative modeling approaches.

- Research Assistant in UROP Program, Department of Human Development

Aug. 2015 – May 2018

Advisor: Dr. Kimberly R. Kelly, Child Language Interactions & Memory Lab

- Conducted grounded theory analysis of mother–child narrative recall interactions, focusing on communication dynamics and memory construction.

GRANTS AND FELLOWSHIPS

University of Michigan, Ann Arbor

- Rackham Predoctoral Fellowship (\$62,748) 2026

- Awarded full tuition, fees, and stipends for three academic terms. The Rackham Predoctoral Fellowship is the most prestigious award given by Rackham Graduate School and supports outstanding doctoral students working on dissertations that are unusually creative, ambitious, and impactful.

- Conference Travel Grant (\$900) 2026

- Awarded by the University of Michigan Rackham Graduate School to support presentation of research at the American Psychological Association Research Convention.

- Scholarship for Underrepresented Professionals in Educational Measurement (\$3,000) 2025

- Selected as a fellow and awarded by the Center for Measurement Justice to support attendance and participation in the National Council on Measurement in Education (NCME) annual meeting.

- Rackham PACE Fellowship for Graduate Student–Faculty Mentor Pairs (\$20,916) 2025

- Awarded full tuition, fees, and stipends for one academic terms and \$2,500. The PACE Fellowship is awarded by Rackham Graduate School to support student-faculty mentor pairs who foster a positive program climate and support student success.

- Graduate Student Research Grant (\$1,500)**
 - Awarded by the University of Michigan Rackham Graduate School to support pre-candidacy research activities.

2025
- Conference Travel Grant (\$900)**
 - Awarded by the University of Michigan Rackham Graduate School to support presentation of research at ASPIRE, the Human Factors & Ergonomics Society International Annual Meeting.

2025
- Conference Travel Grant (\$900)**
 - Awarded by the University of Michigan Rackham Graduate School to support presentation of research at the International Symposium on Human Factors & Ergonomics in Healthcare.

2024
- DEI Mini Grant (\$3,000)**
 - Awarded by the University of Michigan Department of Industrial & Operations Engineering through HFES to design and implement three departmental DEI-focused events in collaboration with the UM INFORMS student chapter.

2024

HONORS AND AWARDS

University of Michigan, Ann Arbor

- Research Award, IOE-ISyE-MS&E Rising Stars Workshop**
 - One of 29 selected to present a poster at the Rising Stars Workshop for promising early-career researchers in Industrial Engineering and Operations Research; recognized with a People's Choice Award.

2026
- ¡Lánzate! Higher Education Travel Award**
 - Awarded by the Hispanic Association of Colleges & Universities (HACU) to support academic travel, including four round-trip flights.

2025
- NSF GRFP Honorable Mention**
 - Awarded Honorable Mention by the National Science Foundation Graduate Research Fellowship Program (NSF GRFP), a highly competitive national fellowship program with a reduced number of awards in 2025.

2025

Cedars-Sinai Medical Center, Los Angeles, CA

- Best Oral Presentation**
 - Sole recipient, awarded by the Society of Surgical Ergonomics (\$500) at the Inagural Symposium of the Society of Surgical Ergonomics.

2022

California State University, Long Beach

- Outstanding Senior Award**
 - Awarded by the Department of Psychology to students in the top 5% of the graduating class.

2020
- Regional Research Award**
 - Awarded by the Western Psychological Association at the 99th Annual Convention for research presented through Psi Chi, the International Honor Society in Psychology.

2019
- Internship Essay Contest Winner**
 - Awarded by the College of Liberal Arts (\$500) for top-ranked essay among undergraduate submissions.

2018
- First Place Completed Research Award**
 - Awarded by the Department of Psychology at the Psychology Day Symposium as the sole first-place recipient.

2018
- McNair Scholar**
 - Selected as a scholar in the Ronald E. McNair Post-Baccalaureate Achievement Program.

2018

- Recognized for achieving a semester GPA between 3.50 and 4.00.

PUBLICATIONS

*M indicates mentee author

Published Refereed Journal Articles (9)

1. **Zamudio, J.**, Zhang, Q.^M, Quinn, M., Fowler, K. E., Saint, S., & Yang, X. J. (2025). "Invert the pyramid, let internists design the job as pilots do a cockpit": The views of general internal medicine physicians on enhancing well-being through human factors engineering. *Journal of Patient Safety*, 21(7 Suppl), S36–S42. <https://doi.org/10.1097/PTS.0000000000001356>
2. Narula, N., Dixon, F., **Zamudio, J.**, Wooldridge, A. R., & Shao, J. (2025). Sharper minds, safer surgeries: Understanding the importance of cognitive ergonomics – A perspective piece from the Society of Surgical Ergonomics. *American Journal of Surgery*, 116521. <https://doi.org/10.1016/j.amjsurg.2025.116521>
3. Cohen, T., Kanji, F., **Zamudio, J.**, Breese, C., Avenido, R., Yoshizawa, C., Bartkiewicz, S., Catchpole, K., & Anger, J. (2024). Rethinking surgical safety: Investigating the impact of gamified training on severe flow disruptions in surgery. *Journal of Patient Safety*, 20(8), 593–598. <https://doi.org/10.1097/PTS.0000000000001279>
4. Cohen, T. N., Kanji, F. F., **Zamudio, J.**, Shouhed, D., Gewertz, B. L., & Sax, H. C. (2023). Why can't we improve turnover time? A systematic review. *World Journal of Surgery*, 48(1), 72–85. <https://doi.org/10.1002/wjs.12015>
5. **Zamudio, J.**, Woodward, J.^M, Kanji, F. F., Anger, J. T., Catchpole, K., & Cohen, T. N. (2023). Demands of surgical teams in robotic-assisted surgery: An assessment of intraoperative workload within different surgical specialties. *The American Journal of Surgery*, 226(3), 365–370. <https://doi.org/10.1016/j.amjsurg.2023.06.010>
6. **Zamudio, J.**, Kanji, F. F., Lusk, C., Shouhed, D., Sanchez, B. R., Catchpole, K., Anger, J. T., & Cohen, T. N. (2023). Identifying workflow disruptions in robotic-assisted bariatric surgery: Elucidating challenges experienced by surgical teams. *Obesity Surgery*, 33, 2083–2089. <https://doi.org/10.1007/s11695-023-06620-4>
7. Cohen, T. N., Anger, J., Kanji, F., **Zamudio, J.**, DeForest, E., Lusk, C., & Catchpole, K. (2022). A novel approach for engagement in team training in high-technology surgery: The robotic-assisted surgery Olympics. *Journal of Patient Safety*, 18(6), 570–577. <https://doi.org/10.1097/PTS.0000000000001056>
8. Franco, D., **Zamudio, J.**, Blevins, K. M., Núñez-Larios, E. A., Ricoy, U. M., Iñiguez, S. D., & Zavala, A. R. (2020). Early-life ketamine exposure attenuates the preference for ethanol in adolescent Sprague-Dawley rats. *Behavioural Brain Research*, 389, 112626. <https://doi.org/10.1016/j.bbr.2020.112626>
9. Kelly, K. R., Ocular, G., **Zamudio, J.**, & Plascencia, J. (2020). "But what about the beginning?" Young children's independent narrative structure and how mothers promote chronological coherence during narrative conversations. *Narrative Inquiry*, 32(2), 424–451. <https://doi.org/10.1075/ni.19082.kel>

Refereed Proceedings Papers (1)

1. **Zamudio, J.**, Catchpole, K., Kanji, F., Anger, J., & Cohen, T. (2022). Using the National Aeronautics Space Administration Task Load Index (NASA-TLX) in surgery: Considerations for use "in the wild." *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 66(1). <https://doi.org/10.1177/1071181322661>

Papers in Preparation (5)

1. **Zamudio, J.**, Braker, A.^M, Quinn, M., Houchens, N., Fowler, K. E., Saint, S., & Yang, X. J. "I felt like an active participant in my own care": A qualitative analysis of patient perceptions of connection in care. (In preparation).
2. **Zamudio, J.**, Braker, A.^M, Quinn, M., Fowler, K. E., Saint, S., & Yang, X. J. Patient perceptions of disconnection from providers as a disruption of connection: "You feel like they don't give a damn" (In preparation).
3. **Zamudio, J.**, Braker, A.^M, Quinn, M., Fowler, K. E., Saint, S., & Yang, X. J. A framework for antecedents and consequences of experiencing connection and disconnection in care. (In preparation).
4. **Zamudio, J.**, Braker, A.^M, Fowler, K. E., Saint, S., & Yang, X. J. The Connection Scale: Development and psychometric analysis of a patient-provider connection scale for adults. (In preparation).
5. Narula, N., **Zamudio, J.**, Wooldridge, A. R., & Shao, J. The impact of poor physical, cognitive, and macro-ergonomics on burnout in surgeons: A systematic review. (In preparation).

Invited Academic Presentations (6)

1. **Zamudio, J.**, Braker, A.^M, Manojlovich, M., Fowler, K., Houchens, N., Saint, S., & Yang, X. J. (2025). Conceptualizing patient-provider connectedness in the inpatient clinical setting: A preliminary analysis. Poster presented at the 11th Annual Center for Healthcare Engineering & Patient Safety (CHEPS) Symposium, Ann Arbor, MI.
2. **Zamudio, J.**, Braker, A.^M, Manojlovich, M., Fowler, K., Houchens, N., Saint, S., & Yang, X. J. (2025). Conceptualizing patient-provider connectedness in the inpatient clinical setting. Poster presented at the IOE Department 70th Anniversary Symposium, Ann Arbor, MI.
3. **Zamudio, J.**, & Yang, X. J. (2025). Finding your fit in research. Invited lecture presented at the IOE Undergraduate Research Symposium, Ann Arbor, MI.
4. **Zamudio, J.**, & Cohen, T. N. (2024). Implementing COMFORT (Cognitive Optimizations and Maneuvers in the Operating Room): Flow disruptions and interventions. Panelist at the 3rd Annual Research Symposium of the Society of Surgical Ergonomics.
5. **Zamudio, J.**, & Yang, X. J. (2024). A human factors approach to enhancing patient-provider connectedness. Poster presented at the 10th Annual CHEPS Symposium, Ann Arbor, MI.
6. **Zamudio, J.** (2024). Direct observation in surgical settings. Invited guest lecture, IOE 431 – Human-Centered Design, University of Michigan.

Conference Lecture Presentations (5)

1. **Zamudio, J.**, Chrouser, K., & Yang, X. J. (2024, March). Exploring recurrent communication as a facilitator of effective teamwork in robotic surgery: A proposed analysis using recurrence quantification analysis. Lecture presented at the *Robotics Summit of the Human Factors and Ergonomics International Symposium in Healthcare*, Chicago, IL.
2. **Zamudio, J.**, Choi, H., Deng, Z., Qiu, S., & Sun, H. (2023, August). Self-driving cars: Semantic image segmentation of street scenes. Lecture presented at the Virtual Showcase of the AI Campus Initiative, Arkansas Tech University.
3. **Zamudio, J.**, Catchpole, K., Kanji, F., Anger, J., & Cohen, T. (2022, October). Using the National Aeronautics Space Administration Task Load Index (NASA-TLX) in surgery: Considerations for use “in the wild.” Lecture presented at the 66th International Annual Symposium on Human Factors and Ergonomics, Atlanta, GA.
4. **Zamudio, J.**, Kanji, F., Anger, J., Catchpole, K., & Cohen, T. (2022, September). Assessing workload in surgical settings among OR team members: Variations among specialties in robotic-assisted surgery. Lecture presented at the Inaugural Research Symposium of the Society of Surgical Ergonomics (Virtual).
5. **Zamudio, J.**, Ocular, G., & Kelly, K. R. (2018, February). Like mother, like child: Variation of sentence subject use in maternal speech and child responses. Lecture presented at the Student Research Competition, California State University, Long Beach.

Conference Poster Presentations (23)

1. **Zamudio, J.**, Braker, A.^M, Quinn, M., Harrod, M., Fowler, K., Saint, S., & Yang, X. J. (2026). “Being believed is everything”: Patient Perspectives of Connection in Care. Abstract accepted to the American Psychological Association (APA) Convention in Washington, DC.
2. **Zamudio, J.**, Braker, A.^M, Quinn, M., Harrod, M., Fowler, K., Saint, S., & Yang, X. J. (2026). “I felt like an active participant in my own care”: Understanding Patient Perspectives of Connection and Disconnection for Clinical System Redesign. Abstract accepted to the International Annual Symposium on Human Factors and Ergonomics, in Reno, NV.
3. **Zamudio, J.**, Braker, A.^M, Quinn, M., Harrod, M., Fowler, K., Saint, S., & Yang, X. J. (2026). “Being believed is everything”: Patient Perspectives of Connection and Disconnection in Care. Poster presented at Latinx Research Week at University of Michigan, Ann Arbor, MI.
4. **Zamudio, J.**, Braker, A.^M, Manojlovich, M., Fowler, K., Houchens, N., Saint, S., & Yang, X. J. (2025). Conceptualizing patient-provider connectedness in the inpatient clinical setting: A preliminary analysis. Poster presented at the 69th International Annual Symposium on Human Factors and Ergonomics, Chicago, IL.
5. **Zamudio, J.**, & Mohsin, N. (2024). University of Michigan Center for Ergonomics Laboratories. Poster presented at the 67th International Annual Symposium on Human Factors and Ergonomics, Phoenix, AZ.
6. Lusk, C., Meilan, J., **Zamudio, J.**, et al. (2024). Bridging human factors researchers and practitioners: Current strategies and future directions. Poster presented at the Human Factors and Ergonomics International Symposium in Healthcare, Chicago, IL.
7. **Zamudio, J.**, Chrouser, K., & Yang, X. J. (2024). Exploring recurrent communication as a facilitator of effective teamwork in robotic surgery. Poster presented at Latinx Research Week at University of Michigan, Ann Arbor, MI.
8. Kanji, F., **Zamudio, J.**, Caron, A., Catchpole, K., Anger, J., & Cohen, T. (2022). A human factors approach to understanding workflow challenges between the integration of the da Vinci Multi-Port (Xi) vs. Single Port (SP) robot. Poster presented at the International Symposium on Human Factors and Ergonomics in Healthcare, Orlando, FL.
9. Rana, N.^M, Kanji, F., **Zamudio, J.**, Gewertz, B., Burch, M., & Cohen, T. (2022). Investigation of setup time in high-technology surgery: Opportunities for improvement. Poster presented at the International Symposium on

- Human Factors and Ergonomics in Healthcare, Orlando, FL.
10. Gupta, M.^M, Kanji, F., **Zamudio, J.**, Gewertz, B., & Cohen, T. (2022). Identifying contributing factors to sentinel events in the labor and delivery unit: A human factors approach. Poster presented at the International Symposium on Human Factors and Ergonomics in Healthcare, Orlando, FL.
 11. Cohen, T., Anger, J., Kanji, F., **Zamudio, J.**, Lusk, C., et al. (2022). A novel approach for engagement in team training in high-technology surgery: The robotic-assisted surgery Olympics. Podium presentation at the International Symposium on Human Factors and Ergonomics in Healthcare, Orlando, FL.
 12. Del Gaizo, J., Catchpole, K., Anger, J., Cohen, T., Kanji, F., Lusk, C., **Zamudio, J.**, et al. (2022). A computational model of dynamic mechanisms of failure in surgery. Podium presentation at the International Symposium on Human Factors and Ergonomics in Healthcare, Orlando, FL.
 13. Cohen, T., Avenido, R., Anger, J., Yoshizawa, C., Bartkiewicz, S., Kanji, F., **Zamudio, J.**, et al. (2022). A novel approach for engagement in team training in high-technology surgery: The robotic-assisted surgery Olympics. Poster presented at the Cedars-Sinai Medical Center Nursing Conference, Los Angeles, CA.
 14. **Zamudio, J.**, & Hancock, G. (2020). A technical overview of heart rate variability as an objective measure of cognitive workload. Abstract accepted to the Department of Defense Human Factors Engineering Technical Advisory Group Meeting #74.
 15. Franco, D., **Zamudio, J.**, & Zavala, A. (2019). Early-life ketamine exposure attenuates the rewarding properties of ethanol in adolescent Sprague-Dawley rats. Poster presented at the Society for Neuroscience Annual Meeting, Chicago, IL.
 16. **Zamudio, J.**, & Hancock, G. (2019). A technical overview of heart rate variability as an objective measure of cognitive workload. Poster presented at the McNair Summer Research Internship Program Symposium, California State University, Long Beach.
 17. Franco, D., **Zamudio, J.**, & Zavala, A. (2019). Early-life ketamine administration reverses the aversive properties of ethanol in male adolescent rats. Poster presented at the 99th Western Psychological Association Annual Conference, Pasadena, CA.
 18. Franco, D., **Zamudio, J.**, & Zavala, A. (2019). Functional consequences of juvenile ketamine administration on the rewarding effects of ethanol in male and female adolescent rats. Poster presented at the CSU Program for Education & Research in Biotechnology Annual Conference, Anaheim, CA.
 19. **Zamudio, J.**, Martin, J., Allen, D., & Span, S. A. (2018). The relation among ADHD symptoms, alcohol consumption, and distress tolerance in undergraduates. Poster presented at the 98th Western Psychological Association Annual Conference, Portland, OR.
 20. **Zamudio, J.**, Ocular, G., & Kelly, K. R. (2018). Like mother, like child: Variation of sentence subject use in maternal speech and child responses. Poster presented at Psychology Day Research Fair, California State University, Long Beach.
 21. Kelly, K. R., Ocular, G., Plascencia, J., & **Zamudio, J.** (2017). "But what about the beginning?" Maternal strategies for scaffolding children's chronological sequencing in narrative conversations. Poster presented at World Anthropology Day Research Fair, California State University, Long Beach.
 22. **Zamudio, J.**, & Borrego, J. J. (2017). Mother-child co-construction: The effects of attachment and maternal presence on self-referencing in narratives. Poster presented at the Undergraduate Research Opportunity Program Symposium, California State University, Long Beach.
 23. **Zamudio, J.**, & Vang, J. (2016). Mothers and children co-constructing and telling structurally coherent personal narratives. Poster presented at the Undergraduate Research Opportunity Program Symposium, California State University, Long Beach.

Invention Disclosure

The Robotic-Assisted Surgery Olympics

- Training intervention designed to improve technical and non-technical skills of surgical teams in robotic-assisted surgery. Disclosed and licensed via Cedars-Sinai Medical Center Office of Technology Transfer under a Creative Commons license.

TEACHING AND MENTORING

Teaching Experience

- | | |
|---|--------------------|
| <p>- Instructor, IOE 431 – Human-Centered Design</p> <ul style="list-style-type: none"> • Guest lecture: "Direct Observation in Surgical Settings," University of Michigan. | <p>2024</p> |
| <p>- Instructor, Summer Research Opportunity Program (SROP)</p> <ul style="list-style-type: none"> • Workshop: "Introduction to R Programming." | <p>2024</p> |

- **Instructor, Rackham Aspiring Fellows (RAF) Program** **2024**
 - Workshop: Research Plan Development; Mentor–Mentee Relationships; Leveraging Social Media; Research Journey-Mapping.
- **Facilitator, Discover Engineering Day, University of Michigan** **2023**
 - Co-facilitated process improvement warehouse activity for 8th–10th grade students.
- **Facilitator, The Robotic-Assisted Surgery Olympics** **2022 – 2023**
 - Co-designed and co-facilitated workflow and safety gameified training at Cedars-Sinai Medical Center and the Medical University of South Carolina to over 40 surgical staff members.

Formal Mentoring Experience

- **Research Supervisor, University of Michigan** **2024 – Present**
 - Mentored one post-doctoral researcher (Qiaoning "Carol" Chang) and one graduate student (Jinjin "Jessica" Cai) on individual research projects. Currently mentoring one clinical researcher (Amber Braker) and one undergraduate (Samantha Megdanoff) on individual research projects.
- **Mentor, HFES Mentoring Program** **2025 – Present**
 - Mentoring one early-career student member (Alexander Sipniewski).
- **Mentor, Graduate Application Mentoring Program (GAMP)** **2023 – 2024**
 - Mentored six prospective PhD applicant in Industrial Engineering through graduate application process.
- **Mentor, HFES Mentoring Program** **2023 – 2024**
 - Mentored one early-career industry member (Nohra Murad).
- **Research Supervisor, Cedars-Sinai Medical Center** **2022**
 - Co-mentored three undergraduate students (one social sciences, two pre-med) in individual research projects during their summer internship (Meera Gupta, Madison Brown, and Navin Rana).

Materials Developed

- **Designer, Robotic-Assisted Surgery Olympics Implementation Guide** **2023**
 - Developed an 80-page guide for surgical staff to replicate our multi-site team training intervention: *The Robotic-Assisted Surgery Olympics: A Guide for Hospital Operating Room Management*

Training for University Teaching

- **Graduate Teacher Certificate, University of Michigan** **In Progress**
 - Professional development in evidence-based pedagogy, instructional practice, faculty-mentored teaching, and reflective teaching philosophy development.
- **Emerging Engineering Educators (EMEE) Workshop** **2026**
 - Selected participant in a national faculty development program focused on evidence-based engineering pedagogy, course design, and integration of the entrepreneurial mindset in engineering education.
- **Rackham-CRLT Preparing Future Faculty (PFF)** **2026**
 - Selected participant in a five-week pedagogical training program focused on university teaching, course design, teaching statements, and academic career preparation.
- **IOE, ISyE, MS&E Rising Stars Workshop** **2026**
 - Selected participant in a national faculty development workshop (hosted by the University of Michigan, Georgia Institute of Technology, and Stanford University) focused on academic career preparation, research presentation, mentorship, and professional networking in industrial engineering, operations research, and management science.

- **Promoting Successful Teamwork in STEM Classes (CRLT)** **2025**
 - Completed training on research-based strategies to foster effective teamwork and address common team challenges in STEM courses.
- **An Instructor's Guide to Promoting Student Mental Health & Well-Being (CRLT)** **2025**
 - Completed training on proactive course design, supportive teaching practices, and responding effectively to student distress.

SERVICE

Professional/Academic Societies

- Member, Society of Hispanic Professional Engineers (SHPE) 2023 – Present
- Member, Graduate Society of Women Engineers (GradSWE) 2023 – Present
- Member, Human Factors Transforming Healthcare (HFTH) 2023 – Present
- Member, Human Factors & Ergonomics Society (HFES) 2023 – Present
- Member, Society of Surgical Ergonomics (SSE) 2022 – Present
- Member, Psi Chi International Honor Society in Psychology 2018 – Present

Professional Service and Reviewing (National)

- Reviewer, *Journal of Robotic Surgery* 2025
- Reviewer, *Applied Ergonomics* 2022, 2025
- Program Committee Member, Society of Surgical Ergonomics Symposium 2023 – 2024
- Research Consortium Member, Society of Surgical Ergonomics 2023 – 2024

Service Activities (University-Wide)

- Panelist, Puentes Organization: "How to Be a Graduate Student" Workshop 2025
- Graduate Volunteer, Native American Student Association Visit 2025, 2023
- Panelist, Summer Research Opportunity Program (SROP) Workshop 2024
- Graduate Volunteer, Summer Research Opportunity Program (SROP) 2024
- Graduate Volunteer, Siena Heights & Adrian College McNair Student Visit 2023
- Graduate Volunteer, Explore Michigan Engineering Research & Grad. Education 2023
- Facilitator, Discover Engineering Day Workshop 2023

Service Activities (Departmental)

- Graduate Volunteer, Graduate Application Mentoring Program (GAMP) 2026
- Member, IOE Diversity, Equity, & Inclusion Committee 2023 – Present
- Secretary, Human Factors & Ergonomics Society (HFES) UM Chapter 2024 – 2025
- Member Representing HFES, IOE Student Leadership Board 2023 – 2025
- DEI Advocate, Human Factors & Ergonomics Society (HFES) UM Chapter 2023 – 2024
- Graduate Volunteer, Graduate Application Mentoring Program (GAMP) 2025
- Invited Panelist, First-Year PhD Graduate Student Orientation 2024, 2025
- Invited Panelist, Future Graduate Student Workshop 2023, 2024
- Judge, IOE Undergraduate Research Symposium 2024
- Facilitator, IOE Community Dialogue Seminar Series 2024
- Graduate Volunteer, Graduate Student Recruitment Day 2024

MEDIA COVERAGE

News Articles

- **"Two students awarded the Rackham Predoctoral Fellowship "** **2026**
 - Kirkland, E. *Industrial and Operations Engineering Department News*, January 2026.
- **"Beyond the classroom: Jennifer Zamudio, preserving humanity in health care"** **2026**
 - Kirkland, E. *Industrial and Operations Engineering Department News*, January 2026.
- **"U-M IOE awarded at HFES annual meeting"** **2025**
 - Tamez, J. *Industrial and Operations Engineering Department News*, October 2025.

- “Jennifer Zamudio and X. Jessie Yang awarded PACE Fellowship”**

 - Kirkland, E. *Industrial and Operations Engineering Department News*, June 2025.

2025
- “Three students recognized with NSF GRFP honorable mentions”**

 - Tamez, J. *Industrial and Operations Engineering Department News*, May 2025.

2025
- “U-M IOE earns top HFES honors”**

 - Tamez, J. *Industrial and Operations Engineering Department News*, September 2024.

2024
- “Serious fun: Robot Olympics helps surgical teams hone teamwork skills”**

 - McGhee, K. *MUSC Catalyst News*, January 2023.

2023
- “First-Gen Conference is the first of its kind at CSULB”**

 - Rivero, G. *Long Beach Current*, November 2019.

2019